TED (21) - 3053 Reg. No...............................

REVISION: 2021 Signature………………….

DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY/

MANAGEMENT/COMMERCIAL PRACTICE

AUTOMOBILE ELECTRICAL AND ELECTRONICS SYSTEMS

Time: 3 hours Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module outcome	Cognitive level
1	Define insulator.	M1.01	R
2	Write any one application of SCR.	M1.02	R
3	Write any two battery troubles.	M1.04	R
4	What is the energy conversion in an alternator ?	M2.01	R
5	Write the function of the solenoid switch in the starting system.	M2.04	R

6	Write the function of contact breaker point.	M3.01	R
7	Write the function of the ignition coil.	M3.01	R
8	What is an odometer?	M4.04	R
9	What is a speedometer?	M4.04	R

PART B

II. Answer any eight questions from the following.

(8 x 3 = 24 Marks)

		Module outcome	Cognitive level
1	Write the constructional details of P type semiconductors.	M1.02	U
2	Write a short note about silicon controlled rectifiers.	M1.02	R
3	Write the working principle of an AC generator.	M1.03	U
4	Draw the "delta " connection diagram with full wave rectification diodes used in stator winding of alternator.	M2.02	A
5	Write a short note about axial starter.	M2.04	A

6	Draw series winding type starting motor and label it.	M2.03	R
7	Write a short note about hall effect switches.	M3.03	R
8	Write any three advantages of distributor less electronic ignition systems.	M3.03	U
9	Write a short note about anti theft system.	M4.04	R
10	Prepare a short note about the keyless entry system.	M4.04	U

PART C

III. Answer all questions from the following

(6 x 7 = 42 Marks)

		Module outcome	Cognitive level
1	Explain the working of Lead – Acid battery.	M1.03	U
	OR		
2	Describe the basic operation of the transistor.	M1.02	R
3	Explain the working principle of Lithium – Ion battery.	M1.03	U
	OR		

4	Explain the constructional details of the alternator.	M2.01	U
5	Describe the working of Bendix drive mechanism with a sketch.	M2.04	A
	OR		
6	Summarize the construction of an automotive starting motor.	M2.03	U
7	Explain the constructional features of spark plug.	M3.03	U
	OR		
8	Illustrate the working of distributor type electronic ignition.	M3.02	U
9	Write a short note about 1. Capacitor discharge ignition system (3.5 mark) 2. Waste spark system. (3.5 mark)	M3.02	U
	OR		
10	Write a short note about 1. Daytime running light. (3.5 mark) 2. Blinker lights. (3.5 marks)	M4.04	U

11	Summarize the constructional features of sealed beam headlight.	M4.02	U
OR			
12	Prepare headlight aiming procedure.	M4.04	A